

Muhammad Hammad Irfan

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RESEARCH INTERESTS

Adversarial ML, LLM and agentic AI security, data leakage in retrieval-augmented systems, and trustworthy, explainable, responsible multi-agent AI evaluation.

EDUCATION

National University of Sciences & Technology (NUST) Islamabad, Pakistan
Bachelor of Science in Computer Science (SEECs) Sep 2023 – Present
Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Large Language Models, Cyber Security, Software Engineering, Advanced Database management Systems, HCI & Computer Graphics, Probability & Statistics, Linear Algebra

RESEARCH EXPERIENCE

Aginiti Redteam *Python, LiteLLM, ChromaDB, FastAPI, pytest, PyPI | [GitHub](#)*
DevNeuron, Sunnyvale CA (Remote) – Forward-Deployed AI Engineering Intern, supervised by Dr. Ahmad Faraz Khan
Mar 2026 – Present

- Built an open-source adaptive red-teaming engine for agentic AI and RAG systems that keeps an evolving model of everything it has learned about a target and re-ranks candidate attacks at every step using a weighted scoring formula (expected new information, potential impact, prior failures), rather than following a fixed script, mapping every finding to the OWASP LLM Top 10.
- Implemented four attacks from published security research spanning three threat models: IKEA (ICLR 2026) and SECRET (IEEE TIFS 2026) for RAG knowledge-base reconstruction, the Interrogation Attack (ACM CCS 2025) for membership inference, and SPE-LLM (ICLR 2026) for system-prompt extraction; verified every finding in three steps, a plain-text scan for obvious matches, an LLM check against a specific list of what a real leak looks like, and a final direct comparison against the actual source data to confirm the match.
- Benchmarked the planner against random and fixed-checklist attack selection on a mock enterprise assistant running 8 defenses (access control, output filtering, rate limiting, and more): all three found the same vulnerabilities, but the adaptive planner needed 9.7 attempts on average versus 50.7, and triggered the target's own filters only 0.7 times versus 40.7. Matched NVIDIA's garak, an industry-standard scanner, exactly on every shared category, and confirmed working exploits on two independently-built systems: AnythingLLM and a 19-agent, three-protocol test fleet.

KidneyCare Clinical Assistant *Next.js, TypeScript, PostgreSQL, Supabase, SQL | [GitHub](#)*
SEECs, NUST – AI and Software Developer, Industry Project (with Kyung Hee University Hospital), supervised by Dr. Syed Imran Ali
Jun 2025 – Feb 2026

- Engineered the explainable decision-support system of a rule-based clinical assistant for nephrology built with Kyung Hee University Hospital, translating physician-validated clinical logic into deterministic classification across 33 clinical situations derived from clinical groups, PTH thresholds and trends, corrected calcium, and phosphate levels.
- Designed longitudinal data handling that compares current results against the most recent prior visit and enforces per-test validity windows, then maps patients into one of 33 clinical situations so doctors see recommendation options consistent with similar past cases.
- Validated the rule engine against historical patient records from Kyung Hee University Hospital across all 33 clinical situations, boundary conditions, and edge-case scenarios, confirming concordance between the system's recommendations and actual nephrologist decisions.

S2Cool – Sustainable Cooling Research Consortium *Python, NVIDIA Omniverse USD Composer, Ridge Regression*
NUST SEECs – UKRI Ayrton Challenge Consortium (led by Northumbria University), supervised by Dr. Moazam Fraz and Dr. Naseer Bajwa
Jan 2026 – Present

- Contributing to S2Cool, a £2.8M UKRI Ayrton Challenge consortium led by Northumbria University, building a digital-twin model of a refrigerant-free evaporative cooling system targeting up to 65% lower cooling energy demand in extreme-heat climates.
- Transitioning the system model toward a dual-track behavioral model of the cooling system that combines first-principles heat and mass-transfer equations with data-driven ML surrogate models, trained on roughly 66,000 empirical observations from 11 instrumented test runs on the prototype.
- Training Ridge regression surrogate models per subsystem against sensor data, currently reaching $R^2 \geq 0.93$ and $RMSE < 0.5$ for supply-temperature and thermal-effectiveness prediction.

PROJECTS

StackShadow – Autonomous AI CTO Co-Pilot *2nd Place, Google Build with AI Hackathon 2026 (GDG on Campus, FAST NUCES)*

- Built a Next.js/Supabase multi-agent platform automating infrastructure and cost analysis, with AI agents via Gemini and GitHub APIs to map dependencies, detect vulnerabilities, and optimize tooling spend.

AssanKhata – Multi-Tenant Business Management SaaS

- Independently scoped, built, and deployed a multi-tenant SaaS (Next.js, Supabase) for a real printing business, automating invoicing, order management, payments, and stock tracking with offline-capable workflows, using Google OAuth and Supabase Row-Level Security for organization-level data isolation across multiple independent businesses.

HireMatch – AI-Powered Hiring Match Platform

- Built an applicant-tracking system using PGVector for semantic resume-to-job matching, with an LLM pipeline generating candidate relevance scores and personalized skill-gap analyses.

EduSync – Role-Based University Automation Platform

- Built a full-stack university automation platform (Next.js, Supabase, Prisma ORM) modeling real institutional hierarchies across departments, years, sections, and advisors, with role-based access control for admin, faculty, and student workflows including scheduling, onboarding, and attendance.

LingoMate – AI-Powered English Speaking Companion

- Built a full-stack AI voice-conversation app (React, Node.js, Express) using Azure Speech-to-Text/Text-to-Speech and Gemini, handling real-time browser audio capture and managing latency across the speech-recognition, generation, and synthesis pipeline.

TECHNICAL SKILLS

Research & ML Systems: PyTorch, scikit-learn, NumPy, Pandas, Matplotlib, Seaborn; LLM red-teaming and adversarial evaluation, RAG, embeddings, prompt engineering, multi-agent orchestration and security, agentic AI evaluation

Programming & Systems: Python, TypeScript/JavaScript, SQL; Git, Docker, Linux

Data & AI Infrastructure: LiteLLM, ChromaDB, vector search (PGVector), FastAPI, PostgreSQL/Supabase, Prisma ORM, RBAC & Row-Level Security, OWASP LLM Top 10

Engineering Practice: System Design, CI/CD, Production Deployment

AWARDS & ACHIEVEMENTS

Achieved a merit rank of 284 out of approximately 75,000 applicants in the NUST Entry Test (NET), earning admission to SEECS.

2nd Place, Google Build with AI Hackathon 2026 (GDG on Campus, FAST-NUCES), for StackShadow, an autonomous multi-agent infrastructure-analysis platform.